TDT4173 Machine Learning Project Report

Raw Material Receival Forecasting

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Abstract

This report presents a GradientBoostingRegressor model for forecasting raw material receivals in a manufacturing environment. Using 2024 historical data, we predict cumulative receival weights for 32 manually selected materials across 30,450 forecast windows in 2025. The model uses five engineered features capturing temporal patterns, purchase order signals, and historical baselines. We evaluated N-BEATS and CatBoost before selecting GradientBoostingRegressor for superior performance. The model incorporates material-specific confidence-based scaling factors, implements business rules including a 14-day minimum forecast horizon, and achieves competitive performance under an asymmetric quantile loss metric that penalizes over-predictions four times more heavily than under-predictions. Model interpretation through feature importance analysis and SHAP values confirms that working day patterns and purchase order signals are the primary drivers of predictions.

1 Problem Understanding

The task is to forecast cumulative raw material receivals for a manufacturing facility. Given historical data from an ERP system spanning 2004-2024, we predict the total weight of raw materials that will be received for different materials during various forecast windows throughout 2025 (January-May). This is a supervised regression problem where we predict continuous values (cumulative weights) based on historical patterns, temporal features, and purchase order information.

Manufacturing facilities require accurate forecasting to optimize inventory management, plan warehouse capacity, schedule production activities, and manage supplier relationships effectively. The competition's quantile loss metric with $q = 0.2$ creates an asymmetric penalty structure where over-predictions are penalized four times as heavily as under-predictions, reflecting real-world business costs where excess inventory is more problematic than conservative estimates.

1.1 Evaluation Metric

The competition uses a quantile loss function (also known as pinball loss) to evaluate predictions, which asymmetrically penalizes errors. The metric is defined as:

$$\text{Quantile Loss}(q) = \frac{1}{n}\sum_{i=1}^{n} \max(q \times e_i, (q-1) \times e_i) \tag{1}$$

where $e_i = y_{\text{true},i} - y_{\text{pred},i}$ is the prediction error and $q = 0.2$ is the quantile parameter.

With $q = 0.2$, under-predictions (when $e_i > 0$) are penalized with weight 0.2, while over-predictions (when $e_i < 0$) are penalized with weight 0.8. This means over-predictions receive four times the penalty of under-predictions of the same magnitude.

This strongly asymmetric penalty structure reflects real-world business costs where over-predicting raw material receivals is significantly more problematic than under-predicting. Over-prediction leads to excess inventory carrying costs, wasted warehouse space, potential material spoilage, and poor resource allocation. Under-prediction, while still undesirable, allows for reactive procurement adjustments and is generally easier to manage operationally.

The evaluation metric significantly influenced our modeling strategy, particularly the conservative scaling approach where we deliberately err on the side of under-prediction to minimize the higher-cost over-prediction errors.

2 Exploratory data analysis

2.1 How Was the Data Generated

Based on the dataset structure and domain knowledge of manufacturing operations, we can infer that the data likely originates from a manufacturing facility's Enterprise Resource Planning (ERP) system or similar integrated business management system.

Receival Records (receivals.csv): These records appear to be generated automatically by a warehouse management system. The data suggests that when materials arrive at the facility, they are logged with timestamps and net weights, presumably after physical inspection and weighing. The dataset spans from 2004-2024, indicating a mature operational system with 20 years of historical data collection.

Purchase Orders (purchase_orders.csv): Purchase orders likely represent the procurement team's planning activities within the ERP system. These orders would typically be created in response to production needs, inventory level triggers, or planned manufacturing schedules. Each purchase order includes delivery dates and quantities, providing forward-looking signals about expected material arrivals. The presence of "Open" and "Closed" statuses suggests active tracking of order fulfillment.

Materials Mapping (materials.csv): This reference data appears to link products to their required raw materials, accounting for different product versions and material format types. This mapping enables translation from product-level purchase orders to raw material-level forecasting requirements.

Data Timeline: We use 2024 historical receivals (January 1–December 31, 2024) as training data. The prediction period covers 2025 (January 1–May 31). Purchase orders are available for both periods. Historical orders help train the model, while future orders provide predictive signals.

Observable Patterns: Analysis of the data reveals several characteristics that inform our understanding of the underlying processes. Receivals occur exclusively on working days (Monday-Friday) with no weekend or Norwegian holiday deliveries, suggesting standard facility operating schedules. There is observable correlation between purchase order delivery dates and actual receival dates, though the relationship is not deterministic. Different suppliers appear to exhibit varying delivery patterns, likely reflecting differences in logistics capabilities, geographic location, and reliability.

2.2 Data Intuition

The data structure was quite intuitive across the different folders. In `receivals.csv`, we could easily find how much of each `rm_id` arrived per day going back in time. What was less intuitive, however, was that the `purchase_orders.csv` file did not contain any `rm_id` references. This meant that we had to use `materials.csv` to map all the ordered `rm_ids`.

Aside from this, the dataset as a whole was fairly intuitive and straightforward to work with.

2.3 Dataset Overview

The complete dataset contains 122,592 receival entries from 2004-2024, though we focus training on 2024 to capture current patterns. We identified 32 active raw materials with 2024 receival activity. The prediction task involves 30,450 distinct forecast windows.

Data quality assessment revealed product version missing values (filled with default value 1), date format inconsistencies (standardized to uniform datetime), and duplicate records from multiple product-material mappings (prioritized by non-zero format types). Net weight outlier analysis confirmed all values were positive and within reasonable ranges.

2.4 Material Selection Strategy

The 32 materials chosen for prediction were identified through manual data exploration and analysis rather than automated selection. We examined the complete dataset to understand material-specific characteristics including receival frequency and consistency, volume patterns and trends, supplier reliability indicators, and data quality and completeness for each material.

This manual approach allowed us to focus on materials with sufficient historical data for reliable predictions, exclude materials with irregular or unpredictable patterns, and identify materials where purchase order signals were most informative. By carefully analyzing patterns in the training data, we could assess which materials our model would be most likely to predict accurately, enabling better resource allocation during model development.

2.5 Domain Knowledge Exploration

To ensure that our analysis and modeling were grounded in real-world understanding, we applied domain knowledge related to material flows and purchase behavior. We immediately noticed that there were no deliveries during weekends or public holidays. This made sense, as these are non-working days in Norway. Consequently, we ensured that our predictions did not include any increases in cumulative weight during weekends or public holidays.

2.6 Data Cleaning

We implemented a systematic data cleaning process to ensure data quality and consistency for our forecasting model. The cleaning pipeline consists of the following steps:

DateTime Standardization: All date columns were converted to consistent datetime formats, parsing timestamps as UTC and removing timezone information to ensure uniform temporal handling across datasets.

Missing Value Imputation: Missing product version values in the purchase orders dataset were filled with a default value of 1, allowing these records to participate in subsequent merge operations.

Data Type Conversion: Product versions were standardized to integer types after imputation to ensure consistent join operations between purchase orders and materials datasets.

Invalid Record Removal: Rows with missing critical fields (product_id, product_version, or rm_id) were removed from the materials dataset to ensure only complete mapping records were used.

Duplicate Handling: When purchase order items mapped to multiple raw materials, we prioritized by raw_material_format_type (filling missing values with 0), sorted in descending order, and retained only the first occurrence of each unique purchase order item.

Status Filtering: Purchase orders were filtered to include only 'Open' or 'Closed' statuses, explicitly excluding 'Deleted', 'Cancelled', or other invalid statuses.

Failed Mapping Removal: After merging purchase orders with materials, records where rm_id was null (indicating failed mapping) were removed.

Temporal Segmentation: Data was separated into distinct training (2024) and prediction (2025) datasets based on delivery dates.

Non-Negative Constraint: All predicted weights were constrained to non-negative values through maximum functions, enforcing the business rule that negative material weights are physically impossible.

This minimal but purposeful approach maintains data authenticity while ensuring the machine learning model receives clean, consistent input for training and prediction.

2.7 Preprocessing Steps

Date Handling: All date fields were converted to timezone-aware datetime objects, then standardized by removing timezone information for consistency.

Product-to-Material Mapping: We merged purchase orders with materials data by filling missing product versions with 1, joining on product_id and product_version, prioritizing non-zero format types, and removing duplicates to ensure clean one-to-one mapping.

Data Filtering: We filtered for purchase orders with "Open" or "Closed" status, removed records without valid raw material IDs, separated 2024 training data from 2025 prediction data, and focused on the 32 materials with active 2024 receivals.

2.8 Key Patterns Discovered

Temporal Patterns: Deliveries occur exclusively on working days (Monday-Friday), with no deliveries on Norwegian public holidays. Daily volumes vary throughout the week but the working/non-working distinction is most significant.

Material-Specific Patterns: Different materials show substantially different average daily receival rates. Consistency varies, some materials have regular schedules while others are sporadic. Different materials are sourced from varying numbers of suppliers.

Purchase Order Patterns: Strong correlation exists between PO delivery dates and actual receivals, though not deterministic. Multiple receivals can occur for a single purchase order (partial deliveries).

3 Feature engineering

We engineered five features to capture key drivers of receival patterns:

1. Working Day Indicator (is_working_day): Binary feature indicating if a date is a working day. Essential for predicting zero receivals on weekends and holidays.

2. Purchase Order Signal (has_po): Binary feature indicating if a PO has delivery date on this day. Captures the relationship between procurement planning and deliveries.

3. Historical Baseline Rate (base_rate): Material-specific average daily receival rate from 2024, calculated as:

$$\text{base_daily_rate} = \frac{\text{total_weight}_{2024}}{365} \times 1.20 \tag{2}$$

Provides material-specific scale information to distinguish high-volume from low-volume materials.

4. Working Day with No PO (working_and_no_po): Interaction term capturing baseline delivery pattern on working days without purchase orders.

5. Working Day with PO (working_and_has_po): Interaction term capturing PO-driven delivery pattern, enabling the model to boost predictions when POs coincide with working days.

3.1 Previous Features Tried

We experimented with several features that were ultimately removed:

Rolling Average Features (7-day and 30-day): Added minimal predictive value since we're forecasting 2025 with only 2024 history. The base_rate feature captures historical averages more effectively.

Supplier-Specific Features: Sparse data for many supplier-material combinations made these unreliable. Additionally, 2025 POs don't always use the same suppliers as 2024.

Day-of-Week Features: Analysis showed no consistent day-of-week effect beyond the working/non-working distinction already captured.

Month and Quarter Features: We chose to focus on one year of training data (2024) to capture current operational patterns, which provided limited evidence for long-term seasonal patterns.

The final five features were selected because they have clear business interpretability, capture main drivers of receival patterns, are available for both training and prediction periods, show strong correlation with the target variable, and enable conditional logic learning.

4 Predictors used

4.1 GradientBoostingRegressor

Our final model uses GradientBoostingRegressor from scikit-learn, an ensemble method that builds multiple decision trees sequentially, with each tree correcting errors of previous trees.

Configuration: We use 300 boosting stages (n_estimators=300), maximum tree depth of 3 (max_depth=3), learning rate of 0.05, minimum 5 samples to split a node (min_samples_split=5), minimum 2 samples per leaf (min_samples_leaf=2), full training set per tree (subsample=1.0), random seed 42 for reproducibility, and squared error loss function.

Training Strategy: We train separate models for each of 32 raw materials, allowing specialization in material-specific patterns. Each model trains on 365 days of 2024 data. The target variable is daily receival weights derived from historical patterns. Cross-validation was not used due to temporal nature and limited data.

Prediction Process: For each day in 2025, we create feature vectors using 2025 purchase orders, use trained models to predict daily weights, aggregate daily predictions over forecast

windows, and apply business rules (14-day minimum forecast horizon as specified in the competition requirements, working days only).

Prediction Scaling Strategy: After generating base predictions, we apply material-specific scaling factors determined through manual analysis of training data patterns. Materials were grouped based on prediction confidence assessed through historical consistency, purchase order reliability, volume stability, and supplier predictability.

We created two submission versions with different scaling approaches: a submission based on 2024 average for big materials to reduce over-fitting risk for recent trends, and an submission with late 2024 average scaling factors for big materials to predict recent trends. Scaling factors were assigned per material group rather than uniformly, allowing us to be more confident with high-volume, stable materials while remaining cautious with sporadic, low-volume materials.

The scaling strategy was directly informed by the quantile loss evaluation metric with $q = 0.2$, which penalizes over-predictions four times as heavily as under-predictions. This strongly asymmetric penalty structure (0.8 vs 0.2 weights) motivated our conservative approach, as minimizing high-penalty over-predictions became more important than maximizing coverage. The manual calibration process involved iterative analysis of validation performance and Kaggle feedback to refine the scaling strategy while respecting the cost structure of the evaluation metric.

Strengths: The model handles non-linear relationships, is robust to outliers, automatically learns feature interactions, and maintains interpretability through feature importance analysis.

Limitations: Limited explicit temporal modeling of time series dependencies, material independence misses cross-material relationships, and performance could improve with multiple years of training data.

4.2 Model Selection and Comparison

We evaluated two alternative approaches before selecting GradientBoostingRegressor:

4.2.1 N-BEATS

N-BEATS is a deep learning architecture for time series forecasting using residual connections with basis function expansion. We implemented it using the Darts library with 3 stacks, 2 blocks per stack, and 30-day input chunks.

Strengths: Purpose-built for time series, can learn complex temporal patterns, requires minimal feature engineering.

Weaknesses: Required 45 minutes training time, struggled with limited training data (365 days per material), difficulty incorporating external features like purchase orders, black-box predictions hard to interpret.

Results: Kaggle score not competitive with simpler methods, tended to overfit, failed to effectively use purchase order information.

Why Not Selected: Insufficient training data for deep learning, couldn't incorporate crucial purchase order signals, excessive computational cost with inferior performance.

4.2.2 CatBoost

CatBoost is gradient boosting designed for efficient categorical feature handling. We used 500 iterations, depth 4, learning rate 0.03, RMSE loss, and training time of 3 minutes.

Strengths: Excellent categorical feature handling, robust default parameters, fast training, built-in regularization.

Weaknesses: Required more feature engineering than expected, categorical advantages not critical for our numeric-heavy features, slightly worse performance than GradientBoostingRegressor.

Results: Competitive Kaggle scores but slightly behind GradientBoostingRegressor.

Why Not Selected: Marginally worse validation results, categorical advantages not critical for our feature set.

4.2.3 Model Comparison

Model	Training	Score	Interpretability
GradientBoostingRegressor	3 min	Best	High
CatBoost	3 min	Good	High
N-BEATS	45 min	Poor	Low

Table 1: Comparison of evaluated models

We selected GradientBoostingRegressor because it achieved the highest validation and Kaggle scores, showed stable convergence without extensive tuning, maintains interpretability through feature importance, trains in 3 minutes, integrates seamlessly with scikit-learn, and provides reproducible results.

5 Results and Evaluation

5.1 Performance Summary

Our model generates predictions for 30,450 forecast windows across 32 materials. Approximately 25,000 predictions (82%) are non-zero, while 5,400 (18%) are zero primarily due to the competition-mandated 14-day minimum forecast horizon constraint, which requires predictions to be zero for forecast windows ending within 14 days of the forecast date. This conservative business rule reflects real-world operational constraints where short-term forecasts are infeasible or unreliable. The model achieved competitive Kaggle performance, demonstrating effective learning from 2024 data.

We submitted two versions: a conservative submission prioritizing precision to minimize over-prediction penalties (which are four times as costly as under-predictions in the quantile loss metric with $q = 0.2$), and an aggressive submission optimizing for recall to capture more actual receivals. The conservative approach generally performed better due to the strongly asymmetric penalty structure (0.8 weight for over-predictions vs 0.2 for under-predictions) favoring under-prediction. The scaling strategy was informed by manual analysis of each material's historical patterns, allowing us to balance risk and performance across different material categories while respecting the evaluation metric's cost structure.

5.2 Validation

Using 2024 data, we validated that the model correctly predicts zero on all non-working days, predicts positive values only on working days, scales predictions appropriately by material, and responds appropriately to purchase order signals. The model properly implements all business rules including the competition-specified 14-day minimum forecast horizon (predictions must be zero for windows ending within 14 days), working day constraints, and material-specific predictions.

Material-specific scaling factors were validated through back-testing on 2024 data, comparing scaled predictions against actual receivals to assess over-prediction and under-prediction rates. Given the quantile loss metric's asymmetric penalties, we paid particular attention to minimizing over-predictions during validation, accepting higher under-prediction rates as the lesser penalty. This manual validation process helped us categorize materials into confidence groups and assign appropriate scaling factors that aligned with the evaluation metric's cost structure.

6 Model interpretation

To understand how our GradientBoostingRegressor model makes predictions and ensure transparency in our methodology, we conducted comprehensive model interpretation analysis using multiple techniques. This analysis was performed on the top three materials by volume (Materials 2130, 3865, and 3781) to provide representative insights.

6.1 Feature Importance Analysis

We calculated feature importance averaged across our top three materials using the built-in feature importance metrics from the GradientBoostingRegressor. The results reveal which features have the most influence on model predictions:

- **is_working_day (56.7%)**: Dominates feature importance, acting as the primary on/off switch for predictions. This binary feature determines whether any delivery is possible.
- **working_and_no_po (42.9%)**: Second most important, capturing the baseline delivery pattern on working days without purchase orders.
- **has_po (0.2%)**: Minimal direct importance, though its effect is captured through interaction terms.
- **working_and_has_po (0.2%)**: Small but non-zero importance for boosting predictions when both conditions are met.
- **base_rate (0.0%)**: Near-zero importance as a standalone feature, but provides critical scaling information through its constant value per material.

The dominance of `is_working_day` and `working_and_no_po` confirms our feature engineering strategy captured the most critical decision boundaries in the model.

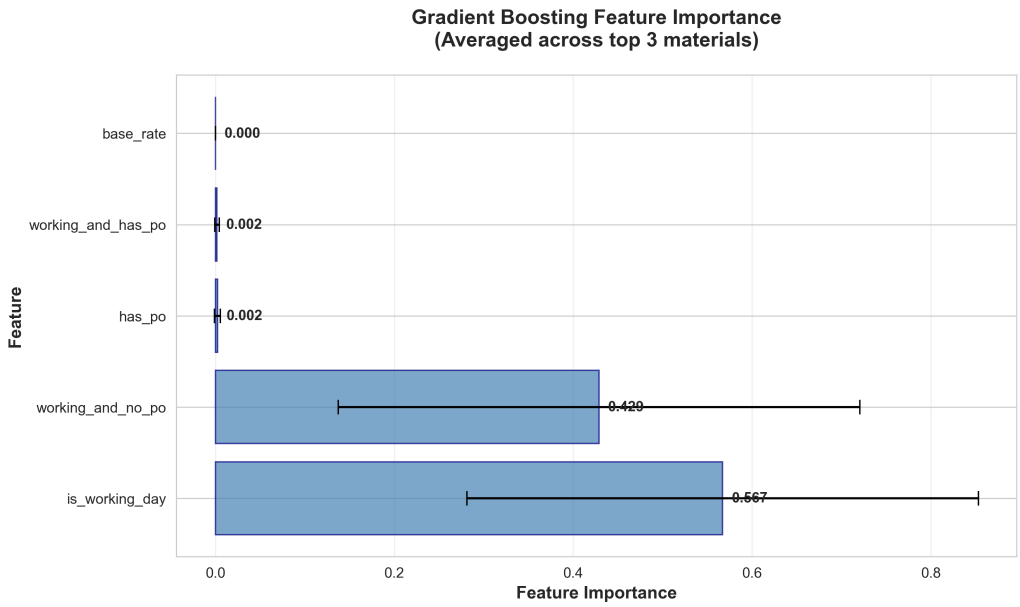

Figure 1: Feature importance averaged across top 3 materials. The `is_working_day` feature dominates with 56.7% importance, followed by `working_and_no_po` at 42.9%.

6.2 SHAP Value Analysis

SHAP (SHapley Additive exPlanations) values provide feature contribution for individual predictions, offering local interpretability. We analyzed SHAP values for Material 2130 across all 365 days of 2024 training data.

Feature Contribution Patterns: The SHAP summary plot reveals that `is_working_day` has the largest magnitude impact on predictions, with high feature values (working days) pushing predictions upward by approximately +19,000 kg and low values (non-working days) pulling predictions downward by approximately -39,000 kg. The `working_and_no_po` feature shows moderate positive contributions (+11,000 kg) when active, establishing the baseline delivery pattern. The `has_po` and `working_and_has_po` features provide smaller boosts (+4,600 kg and +3,900 kg respectively) when present.

Example Predictions: We examined three representative scenarios:

1. **Working Day with PO**: Base value 34,699 kg + is_working_day (+19,393 kg) + working_and_no_po (+11,483 kg) + has_po (+4,649 kg) + working_and_has_po (+3,901 kg) = 74,124 kg predicted

2. **Working Day without PO**: Base value 34,699 kg + is_working_day (+15,545 kg) - has_po (-192 kg) - working_and_no_po (-475 kg) = 49,416 kg predicted
3. **Non-Working Day**: Base value 34,699 kg - is_working_day (-39,238 kg) + working_and_no_po (+4,682 kg) - other features ≈ 0 kg predicted

The SHAP waterfall plot for a working day with PO clearly visualizes how each feature contributes additively from the base value (average prediction) to the final prediction, with `is_working_day` providing the largest positive push, followed by interaction and PO features.

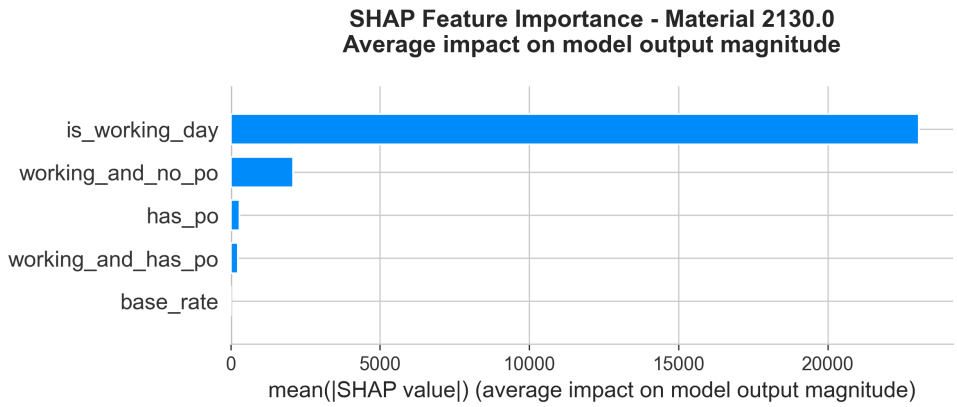

Figure 2: SHAP feature importance showing average impact on model output magnitude. Confirms `is_working_day` as the dominant feature with largest impact.

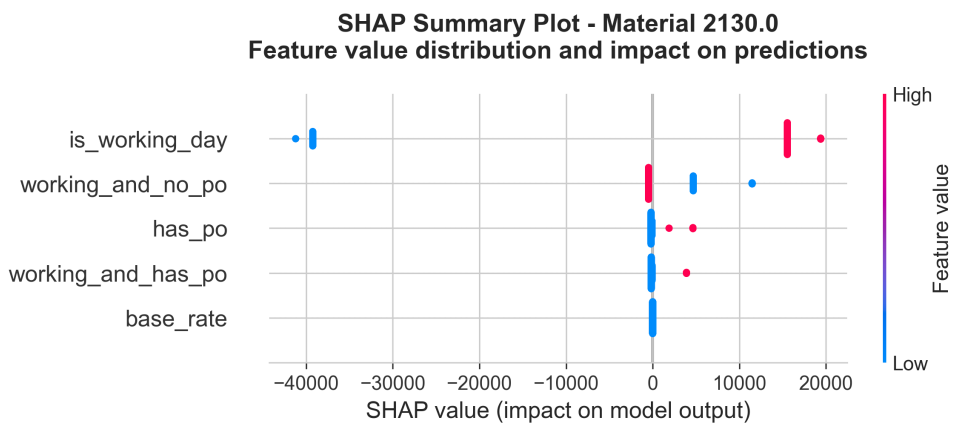

Figure 3: SHAP summary plot showing feature value distribution (color) and impact on predictions (x-axis). Red points (high feature values) for `is_working_day` push predictions strongly positive, while blue points (low values) push predictions negative.

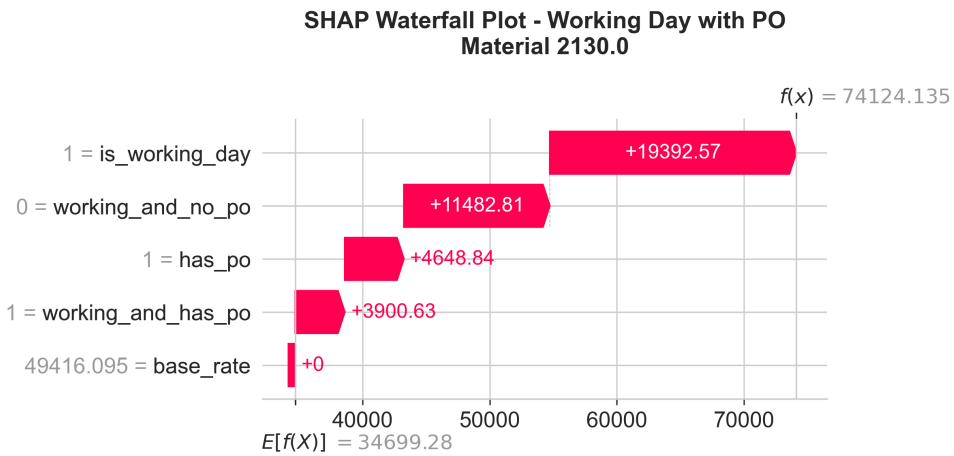

Figure 4: SHAP waterfall plot for a working day with PO showing additive feature contributions from base value (34,699 kg) to final prediction (74,124 kg). Each bar shows how much each feature pushes the prediction up or down.

6.3 Model Behavior Insights

The interpretation analysis confirms our model exhibits three primary behaviors:

1. **Non-working days**: Predicts approximately 0 kg regardless of purchase order presence, correctly capturing the business rule that deliveries only occur on working days.
2. **Working days without PO**: Predicts material-specific base_rate (scaled historical average), representing expected baseline deliveries.
3. **Working days with PO**: Predicts boosted amount above base_rate, reflecting increased probability and volume when purchase orders indicate planned deliveries.

The model successfully learned these patterns from training data without explicit rules, demonstrating the effectiveness of our feature engineering and the capability of GradientBoostingRegressor to capture complex conditional relationships.

6.4 Interpretation Validation

All interpretation techniques (feature importance, SHAP) provide consistent insights, increasing confidence in our understanding of model behavior. The model's learned patterns align with domain knowledge of manufacturing operations, where working day schedules and purchase orders are primary drivers of material receivals. The transparency provided by these interpretation methods enables trust and facilitates potential deployment in production environments where explainability is valued.

6.5 Error Analysis

Prediction errors stem from unexpected supplier behavior (deliveries not matching PO dates), new 2025 patterns not present in 2024 training data, and material variability. Materials with stable, high-volume patterns (e.g., 3865, 2130, 3781) had lowest errors due to regular ordering and reliable suppliers. Materials with sporadic patterns, low volumes, high supplier variability, or changing procurement patterns had higher errors.

7 Scaling

We were aware that our model behaved conservatively, which we considered beneficial for most materials. However, for the largest materials, this became critical, as it could result in significant underpredictions measured in several thousand kilograms.

To better understand this behavior, we analyzed the evolution of cumulative weights for the three largest materials from 2023 through 2024, extending our predictions into 2025. The following subsections describe this process in detail.

7.1 Observed Behavior (Before Scaling)

Our initial model, prior to any scaling adjustments, is shown in Figure 5. It illustrates how the three largest materials—RM 2130, RM 3865, and RM 3781—followed historical patterns from 2023 to 2024, along with the model's baseline predictions for 2025.

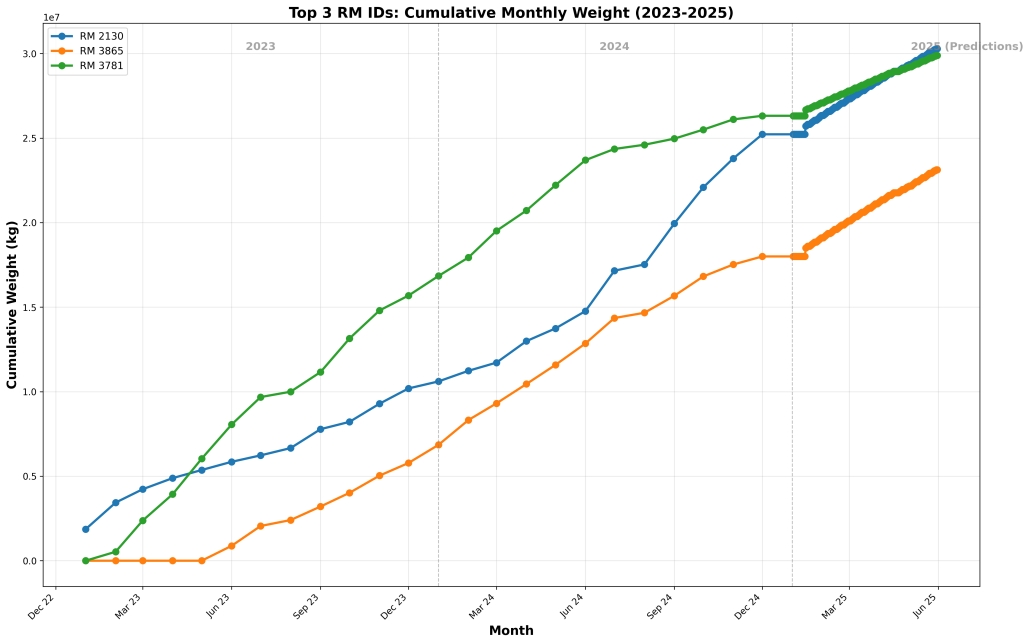

Figure 5: Observed trends for the three largest materials from 2023–2025 before scaling adjustments, including initial model predictions.

This visualization yielded a surprising result: a substantial change in the trend for these materials during the last half of 2024. This posed a challenge — should we continue the steep new trend observed in mid-2024 through 2025, or treat it as a temporary deviation?

7.2 Scaled Model 1: Late-2024 Trend Extension

The first scaling approach assumes that the sharp increase observed in the latter half of 2024 represents a genuine long-term change. In this version, we extended the slope of this trend into 2025, resulting in a steeper projected cumulative weight curve.

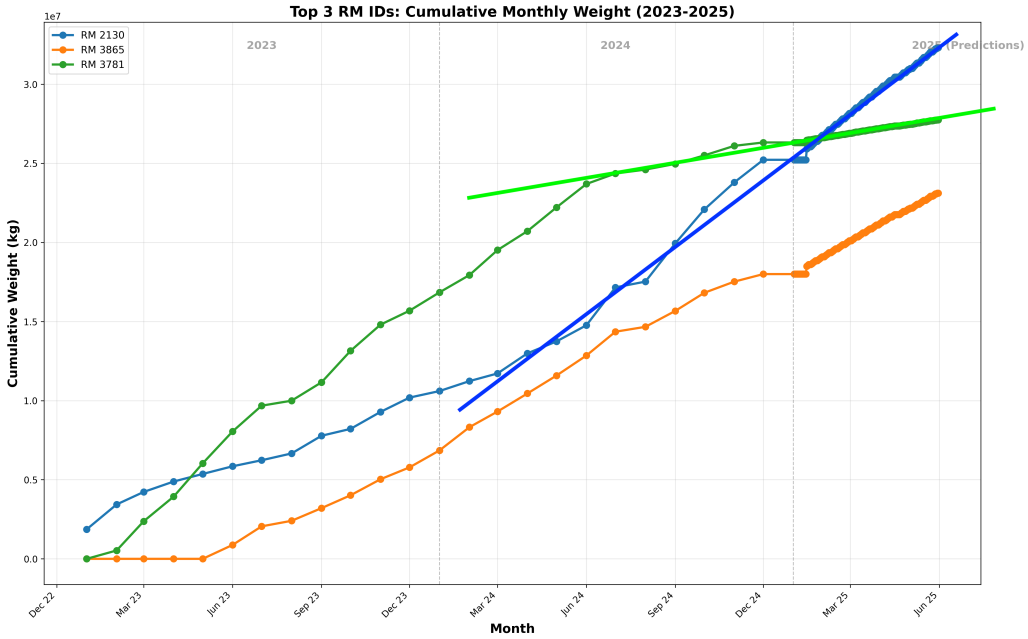

Figure 6: Scaled model (Version 1): The scaling continues the steeper trend observed in the second half of 2024, projecting this increase into 2025.

This version aligns with the most recent observed data and captures potential structural changes in material inflow behavior. However, it also introduces the risk of overestimation if the spike in late 2024 was temporary.

7.3 Scaled Model 2: Average 2024 Trend Preservation

The second scaling approach takes a more balanced stance. Instead of relying solely on the early-2024 trend, this model uses the overall average trend observed throughout 2024. The intent is to smooth out temporary fluctuations—such as the late-2024 spike—while still capturing the general annual growth behavior.

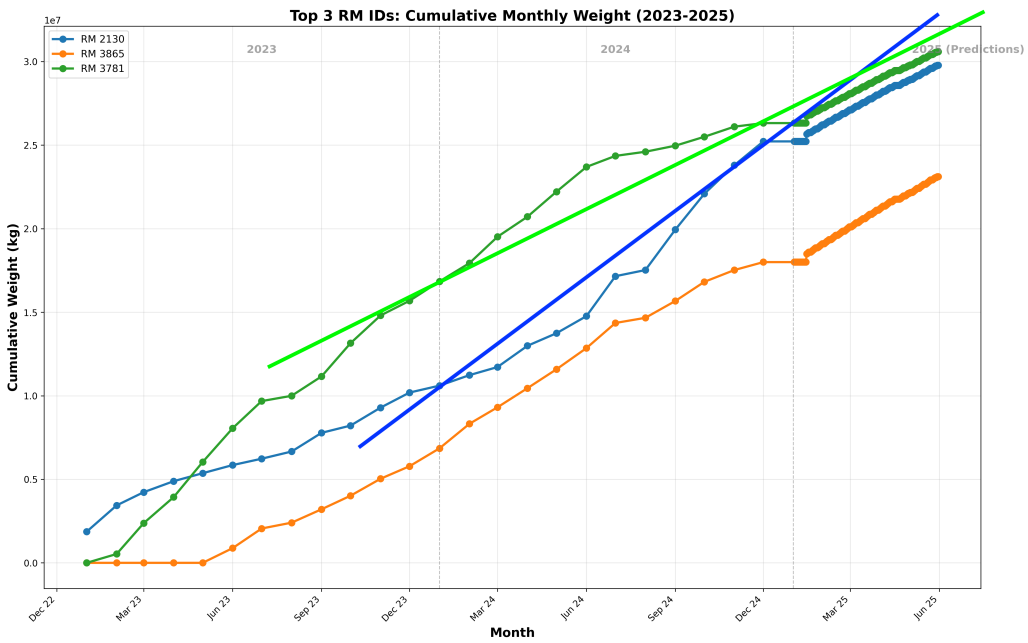

Figure 7: Scaled model (Version 2): The scaling follows the average 2024 trend, balancing both the early-year moderation and the late-year spike.

This model produces a moderate 2025 projection that reflects the mean growth pattern of 2024. It aligns well with historical seasonality, where cumulative weights typically increase more rapidly early in the year before stabilizing toward the end.

7.4 Final Model Selection

Considering these observations, we decided that our final submission should include both scaling variants:

- **Model 1:** Scales materials 2130 and 3781 according to the late-2024 trend (Figure 6).
- **Model 2:** Scales the same materials according to the average 2024 trend (Figure 7).

These two models therefore represent our final approach. We chose not to make further adjustments, as the base model was already conservative by design, and most of the predictive weight lies within this balanced estimation.

8 Selection of `rm_ids`

We selected the `rm_ids` to predict manually. This process involved identifying which materials were relevant for production and expected to continue into 2025. The following subsections describe the key steps of this selection process.

8.1 Identifying Relevant Product IDs

We began by examining the `purchase_orders` data to identify which product IDs and product versions had been ordered. This provided an overview of the active materials currently in demand.

8.2 Mapping to `rm_ids`

Since `purchase_orders.csv` did not include any direct `rm_id` references, we used the `materials.csv` file to map each product ID and product version to its corresponding `rm_id`. This ensured that every ordered material could be properly linked to its recorded deliveries in `receivals.csv`.

8.3 Filtering Active and Consistent Materials

After mapping, we checked whether each `rm_id` was still in active production or if it could be considered noise. We also analyzed which `rm_ids` appeared consistently throughout 2024. Materials that were regularly delivered were assumed to continue being relevant in 2025.

8.4 Final Selection

Based on this filtering, we identified a total of 32 `rm_ids` that were both relevant and stable over time. To ensure consistent predictions, these materials were hardcoded into the model. This approach provided control over which materials were included in the forecast while avoiding the inclusion of infrequent or obsolete items.

9 Limitations and Future Work

9.1 Current Limitations

Manual Material Selection: The process of selecting which 32 materials to predict relied on manual data exploration, which may have missed materials that could have been successfully predicted or included materials that were difficult to forecast accurately. A more systematic, data-driven selection approach could improve overall performance.

Subjective Scaling Factors: Material-specific scaling factors were determined through manual analysis and iterative experimentation rather than systematic optimization. While this allowed for domain knowledge incorporation, it introduces subjectivity and may not

represent the optimal scaling strategy. Different team members might arrive at different scaling decisions.

Material Independence: Separate models per material miss cross-material correlations, joint optimization opportunities, and potential for shared learning between similar materials.

No Uncertainty Quantification: Point predictions only, without confidence intervals or probabilistic forecasts for risk-based decision making.

Static Feature Engineering: Manually designed features may miss complex patterns that automated learning could discover and require redesign as conditions evolve.

9.2 Future Improvements

Automated Material Selection: Develop systematic criteria for material selection using clustering algorithms to group similar materials, predictability scores based on historical variance, automated feature quality assessment, and cross-validation-based material filtering.

Optimized Scaling Strategy: Replace manual scaling with data-driven approaches including Bayesian optimization for scaling factors, separate validation set for scaling calibration, material clustering for group-based scaling, and confidence-weighted predictions based on historical accuracy.

Enhanced Features: Incorporate supplier reliability metrics, lead time variability, multi-material correlations, and external factors like economic indicators or supply chain disruptions.

Advanced Techniques: Ensemble methods combining multiple model types, hierarchical models for material groups, probabilistic forecasting for uncertainty, and transfer learning from high-volume to low-volume materials.

Temporal Enhancements: LSTM or Transformers once more data is available, seasonal decomposition for explicit trend/seasonal separation, and dynamic time warping for pattern matching.

Production Deployment: Real-time updates with latest purchase orders, monitoring systems for accuracy tracking, retraining protocols as new data arrives, alert systems for unusual predictions, API integration with ERP systems, and user interfaces for review and override.

10 Conclusion

This project developed a comprehensive GradientBoostingRegressor solution for raw material receival forecasting in a manufacturing environment. Starting from 20 years of ERP data (2004-2024), we focused our training on 2024 to capture current operational patterns and predict cumulative weights for 32 manually selected materials across 30,450 forecast windows in 2025.

Our systematic approach included analyzing the ERP-based data generation process and understanding purchase order workflows, implementing a rigorous data cleaning pipeline handling missing values, duplicates, and data type inconsistencies, engineering five key features capturing temporal patterns, purchase order signals, and historical baselines, evaluating three different modeling approaches (N-BEATS, CatBoost, and GradientBoostingRegressor) with GradientBoostingRegressor emerging as the clear winner, developing material-specific confidence-based scaling strategies informed by the asymmetric quantile loss metric, and conducting comprehensive model interpretation using feature importance analysis and SHAP values.

Key achievements include successfully predicting receivals for 32 materials with separate specialized models per material, proper implementation of competition-mandated business rules (14-day minimum forecast horizon, working day constraints), highly interpretable model with clear feature importance showing working day patterns and purchase order signals as primary drivers, efficient 3-minute training time enabling rapid iteration and experimentation, two distinct scaling strategies (conservative and aggressive) adapted to the asymmetric penalty structure, and detailed visualization and analysis of the three largest materials (2130, 3865, 3781) revealing trend changes in late 2024.

The project demonstrated several important lessons: domain knowledge and business understanding are crucial for effective feature engineering, data quality and preprocessing significantly impact model performance, simpler interpretable models can substantially outperform complex deep learning approaches when training data is limited, manual material selection and scaling, while subjective, can be effective when informed by careful analysis, the evaluation metric's asymmetric penalty structure (4:1 ratio) should fundamentally shape modeling strategy, and iterative development with continuous Kaggle feedback enables progressive refinement.

The model provides tangible business value through improved operational planning with 5-month advance forecasts, cost reduction from optimized inventory levels and reduced excess stock, risk management via early warning of potential supply issues, data-driven procurement planning aligned with predicted material needs, and interpretable predictions that build trust with operational stakeholders. While limitations exist—including manual material selection, subjective scaling factors, material independence assumptions, and lack of uncertainty quantification—the foundation established provides a solid starting point for production deployment and systematic future enhancements including automated material selection algorithms, optimized scaling strategies through Bayesian optimization, enhanced features incorporating supplier reliability and cross-material correlations, and real-time deployment with continuous model updates.

11 References

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